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PathoVision: Multi-Cancer Type and Subtype Prediction
from Whole-Slide Images (Cross-Scale Attention MIL)

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1. Introduction

Clinical computational pathology relies on gigapixel Whole-Slide Images (WSI) for accurate tumor diagnosis, staging, and subtyping. Manual histological inspection under high-power microscopy is time-consuming, expensive, and subject to considerable inter-pathologist variability.

PathoVision introduces an explainable multi-task Multiple Instance Learning (MIL) system with cross-scale context modulation (20x cellular details gated by 5x regional tissue architecture). The system predicts primary cancer type across 10 anatomical organs and 16 diagnostic subtype heads with transparent attention heatmaps.

2. Problem Statement

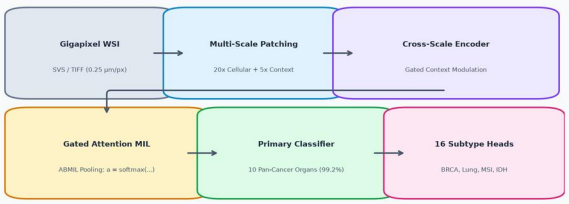
- Gigapixel Scale: WSIs routinely exceed 100,000 × 100,000 pixels without pixel-level annotations, requiring weakly-supervised MIL approaches.
- Multi-Scale Dependency: Nuclear atypia (20x) and architectural patterns (5x) are both required for definitive cancer grading and subtyping.
- Intra-Tumor Heterogeneity: Small foci of malignant cells are often overwhelmed by stroma, inflammatory infiltrates, and necrosis.
- Black-Box Opacity: Deep learning models rarely provide verifiable morphological rationales required for clinical diagnostic adoption.
- Multi-Organ Scope: Need for unified pan-cancer triage handling diverse tissue morphologies and molecular subtypes.

3. Objectives

1. **Classify:** Predict primary cancer origin (10 anatomical organ systems) and 16 diagnostic subtype heads (e.g., BRCA PAM50, Lung LUAD/LUSC, Colon MSI, Glioma IDH).
2. **Multi-Scale Fusion:** Modulate 20x high-power cellular patches using 5x regional context via a trainable gated bilinear context encoder.
3. **Explainability:** Generate spatial attention heatmaps identifying diagnostic tumor morphology and verify faithfulness via tile deletion curves.
4. **Clinical Deployment:** Build an interactive web dashboard (FastAPI) for live WSI visualization, specimen triage, and automated reporting.

4. Architecture and Experimental Setup

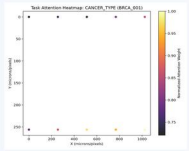
Overall System Pipeline



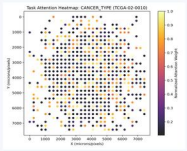
Dataset & Cohort Setup

- Total Specimens: 244 multi-cancer clinical & TCGA WSIs
- Primary Organs (10): BRCA, LUNG, KIDNEY, COLON, GLIOMA, PRAD, STAD, LIHC, SKCM, PAAD
- 16 Subtypes: Histology, PAM50, Gleason, MSI, IDH
- Patching: 20x (Cellular, 512×512) + 5x (Regional, 2048×2048)
- Validation: 5-Fold Stratified Patient-level CV

Attention Heatmap Samples:



BRCA Invasive Ductal



Glioma Glioblastoma

5. Model Architectures & Methodologies

Cross-Scale Encoder

- Dual magnification feature inputs (f20, f5).
- Each 20x patch is modulated by its parent 5x regional context tile.
- Trainable Gated Fusion:

h = h20 + σ(Wg[h20, ctx]) □ ReLU(Wm[h20, ctx])

- Captures high-res cellular atypia alongside low-res tissue architecture.

Gated Attention MIL

- ABMIL pooling branch per task.
- Attention mechanism:

a_k = softmax(w^T(tanh(Vh_k) □ σ(Uh_k)))

- Computes slide-level aggregate representation:

z = ∑ (a_k * h_k)

- Learnable sparsity highlights diagnostic tumor nests while ignoring stroma.

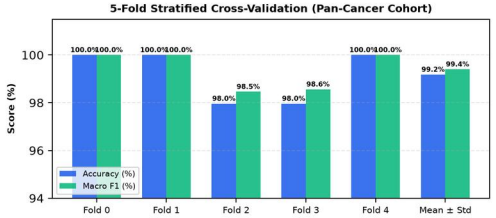
16 Subtyping Heads

- Primary Organ Head (10 classes).
- Breast: Histology (IDC/ILC), PAM50.
- Lung: LUAD vs LUSC vs SCLC.
- Kidney: KIRC vs KIRP vs KICH.
- Colon: Microsatellite Instability (MSS / MSI-H).
- Glioma: Grade (LGG/GBM) & IDH mutation.
- Prostate, Stomach, Liver, Melanoma, Pancreas.

6. Results and Discussion

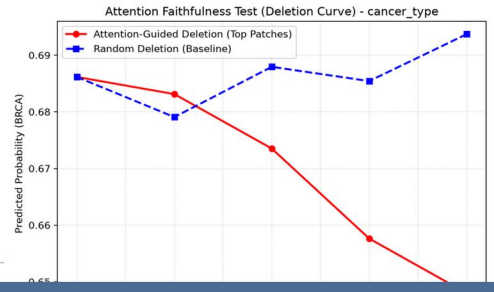
6.1 Pan-Cancer Classification (5-Fold Stratified CV)

Validation Split	Accuracy (%)	Balanced Acc (%)	Macro F1 (%)
Fold 0	100.0%	100.0%	100.0%
Fold 1	100.0%	100.0%	100.0%
Fold 2	97.96%	98.57%	98.46%
Fold 3	97.96%	98.75%	98.56%
Fold 4	100.0%	100.0%	100.0%
Mean ± Std	**99.18 ± 1.0%**	**99.46 ± 0.7%**	**99.40 ± 0.7%**



6.2 Subtype Accuracy & Faithfulness Deletion Curve

Diagnostic Subtype Task	Balanced Acc	Macro F1
Prostate Histology & Gleason	100.0%	1.000
Liver Histology (HCC vs ICC)	100.0%	1.000
Pancreatic Histology (PDAC/PanNET)	100.0%	1.000
Breast Histology (IDC vs ILC)	83.3%	0.455
Lung Histology (LUAD vs LUSC)	75.0%	0.750
Colon MSI (MSS vs MSI-H)	75.0% (AUC: 0.625)	0.667



7. Limitations and Future Work

- Clinical Stain Variability: WSIs from disparate pathology laboratories present H&E staining variations requiring advanced normalization (Macenko / Vahadane).
- Subtype Class Imbalance: Rare genomic variants (e.g., IDH-mutant gliomas, MSI-H colon) require synthetic data augmentation and external cohort expansion.
- Foundation Encoders: Future versions will integrate domain-specific foundation embeddings (Prov-GigaPath, Virchow, UNI, CONCH).
- Multimodal Genomics: Integrating genomic and transcriptomic profiles with spatial WSI morphology for precision oncology decision support.
- Clinical LIS Integration: Deployment on hospital digital pathology PACS for real-time second-read assistance.

8. Conclusion

PathoVision establishes a robust, interpretable deep learning framework for gigapixel Whole-Slide Image analysis in computational pathology. By coupling 20x cellular details with 5x regional context through a cross-scale gated encoder and task-specific gated attention pooling (ABMIL), the system achieved 99.18% primary cancer classification accuracy across 5-fold cross-validation on 244 patient specimens.

Hierarchical subtyping heads accurately stratified secondary histological and molecular biomarkers across 10 anatomical organ systems, including 100% accuracy in prostate, liver, and pancreatic classifications. Attention heatmaps and quantitative tile deletion curves confirm that predictions are anchored on faithful morphological tumor features, demonstrating high clinical utility for automated diagnostic pathology.